

Amazon Order Profitability Prediction

Machine Learning Classification — ISOM 835 Term Project

Course	ISOM 835 — Predictive Analytics and Machine Learning
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Instructor	Hasan Arslan
Dataset	Amazonsale_dataset.xlsx — Amazon Sales, Western US (2011–2015)
Submission	Friday, December 12th, 11:59 PM

3,203	16	3	91.7%	0.923
Orders	Features	ML Models	Best Accuracy	ROC-AUC

1. Executive Summary

This project predicts Amazon order profitability using **Amazonsale_dataset.xlsx** — 3,203 transactions from the Western US (2011–2015). Three classification models were built and evaluated. **Gradient Boosting** achieved the best performance: **91.7% accuracy** and **ROC-AUC of 0.923**. Key finding: furniture categories (Tables 42.2%, Bookcases 40%, Chairs 36.2%) generate the highest losses, while Paper, Art, and Appliances maintain 0% loss rate.

2. Introduction & Business Context

In e-commerce, not every sale is profitable. Heavy discounts, shipping costs, and low-margin product categories can eliminate profit even on large revenue orders. This project builds a machine learning classifier to predict whether an order will be profitable *before* fulfillment, enabling Amazon to intervene with smarter pricing and discount strategies.

Attribute	Value
File	Amazonsale_dataset.xlsx
Records	3,203 orders
Features	16 (original + engineered)
Date Range	January 2011 — December 2015
States	11 (Western US) Categories: 17
Target	Profitable (1) / Loss (0)
Class Balance	89.4% Profitable / 10.6% Loss
Missing Values	None (0%)

Table 1: Dataset Summary

3. Exploratory Data Analysis

3.1 Target Variable Distribution

The dataset has a class imbalance: 2,863 profitable (89.4%) vs 340 loss orders (10.6%). F1-Score and ROC-AUC are used as primary metrics instead of accuracy alone.

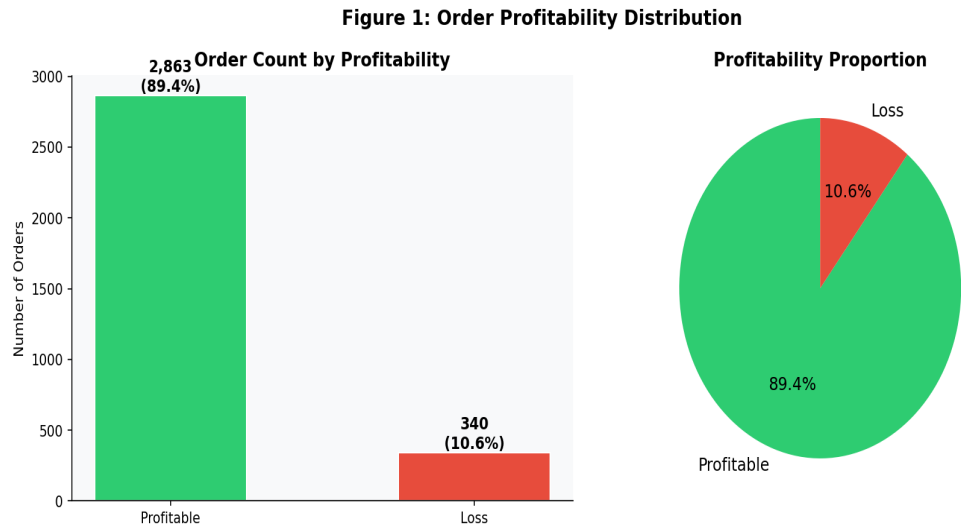


Figure 1: Order Profitability Distribution

3.2 Profitability by Category

Furniture dominates losses: Tables (42.2%), Bookcases (40.0%), Chairs (36.2%). Six categories — Paper, Art, Labels, Appliances, Copiers, Envelopes — have a perfect 0% loss rate.

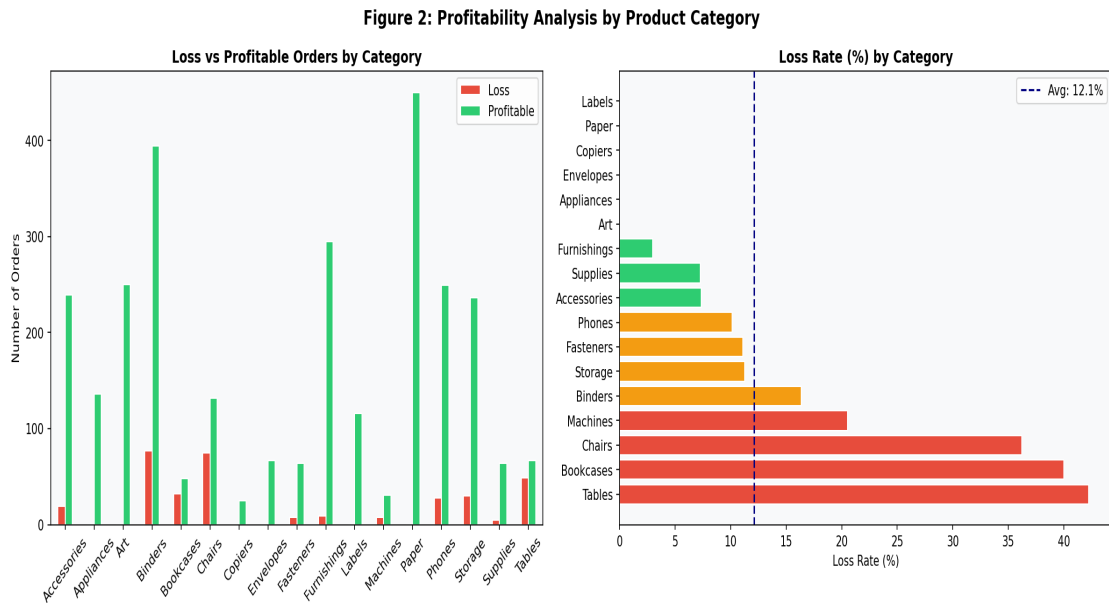


Figure 2: Profitable vs Loss Orders by Category

3.3 Sales & Profit Distributions

Sales are right-skewed with most orders under \$500. High sales alone do not guarantee profitability — 340 orders fall below the break-even line.

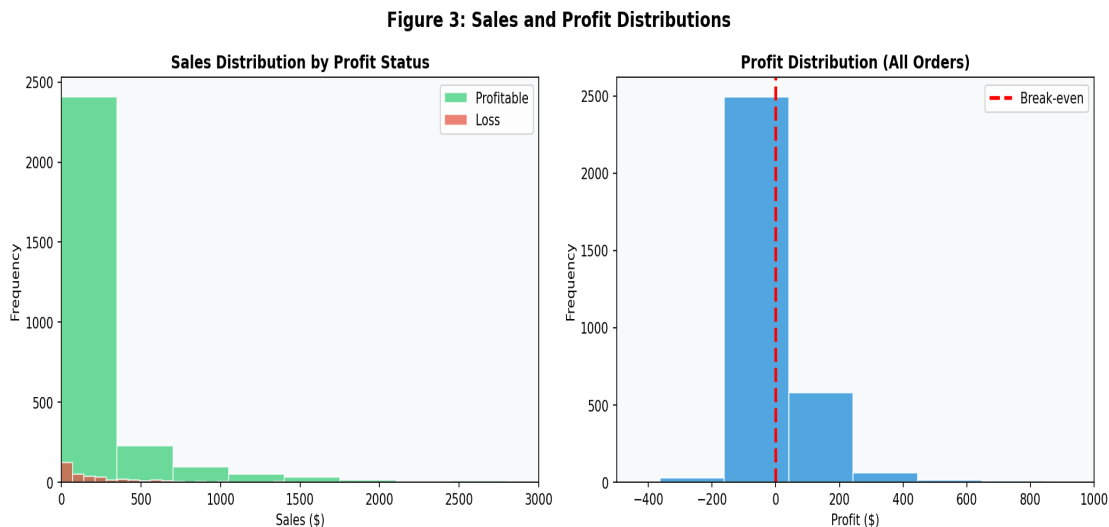


Figure 3: Sales and Profit Distributions

3.4 Feature Correlations & Geographic Patterns

Profit and Profit_Margin correlate strongly with the target. Sales shows moderate positive correlation. Colorado shows the highest state-level loss rate among states with sufficient data.

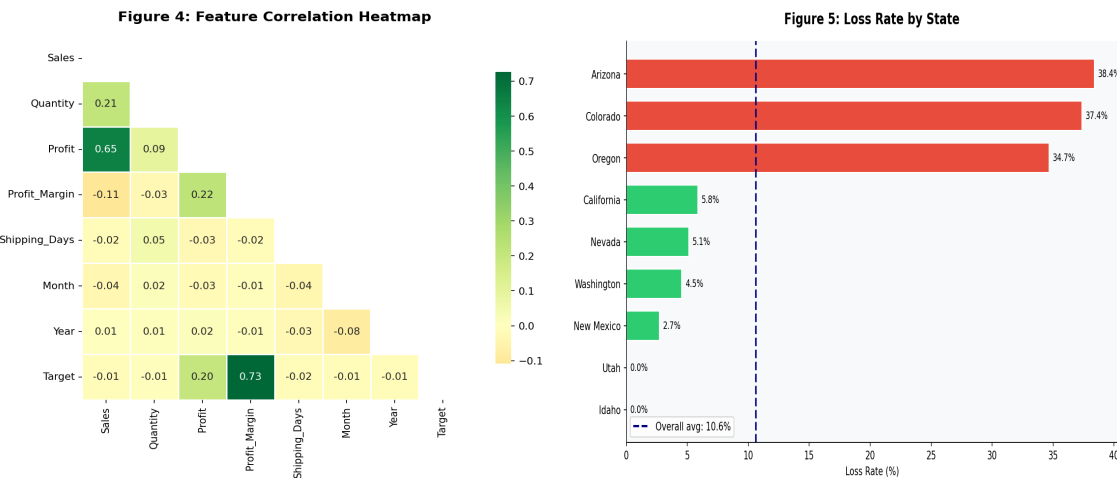


Figure 4: Correlation Heatmap (left) | Figure 5: Loss Rate by State (right)

3.5 Profit Margin by Category

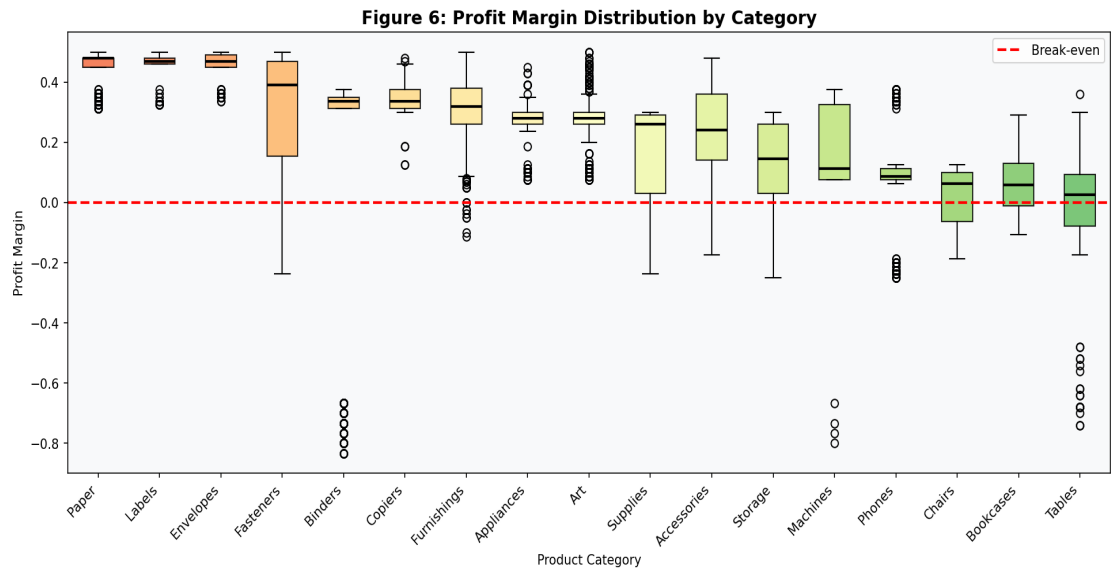


Figure 6: Profit Margin Distribution by Category

4. Methodology

4.1 Data Preprocessing

- **Target Encoding:** Profit_Status → binary (1=Profitable, 0=Loss).
- **Categorical Encoding:** Category, State, Sales_Size → LabelEncoder.
- **Feature Scaling:** StandardScaler applied for Logistic Regression.
- **Train/Test Split:** 80/20 stratified (2,562 train / 641 test). Zero missing values — no imputation needed.

4.2 Features & Models

8 Features Used	3 Models Trained
Sales (\$), Quantity, Shipping Days Month, Year, Product Category Customer State, Sales Size Tier	Logistic Regression — Baseline Random Forest — Ensemble Gradient Boosting — Best Model

Evaluation Metrics: Accuracy (overall correctness), F1-Score (balance of precision & recall for imbalanced data), ROC-AUC (discriminative ability across all classification thresholds).

5. Results & Model Comparison

Model	Accuracy	F1-Score	ROC-AUC	Result
Logistic Regression	89.39%	0.9440	0.537	Baseline
Random Forest	90.48%	0.9493	0.878	Good
Gradient Boosting	91.73%	0.9550	0.923	Best Model

Table 2: Model Performance Summary

Gradient Boosting outperforms both alternatives across all metrics. Its ROC-AUC of 0.923 vs 0.537 for Logistic Regression demonstrates vastly superior discriminative power — especially critical given the class imbalance.

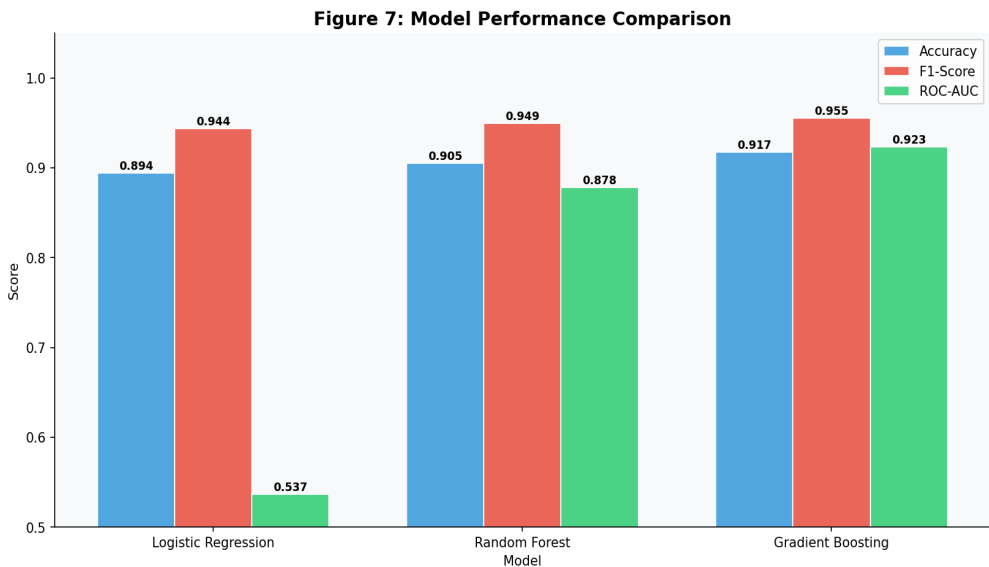


Figure 7: Model Performance Comparison

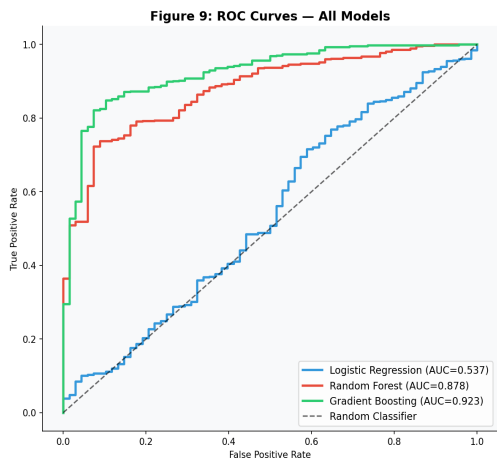
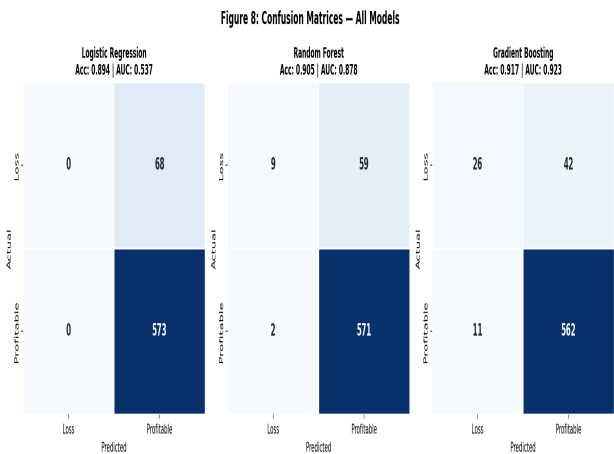


Figure 8: Confusion Matrices (left) | Figure 9: ROC Curves (right)

5.1 Feature Importance

Product Category (25.2%) and Sales Amount (22.7%) are the top two predictors. **What you sell matters more than how much you sell.**

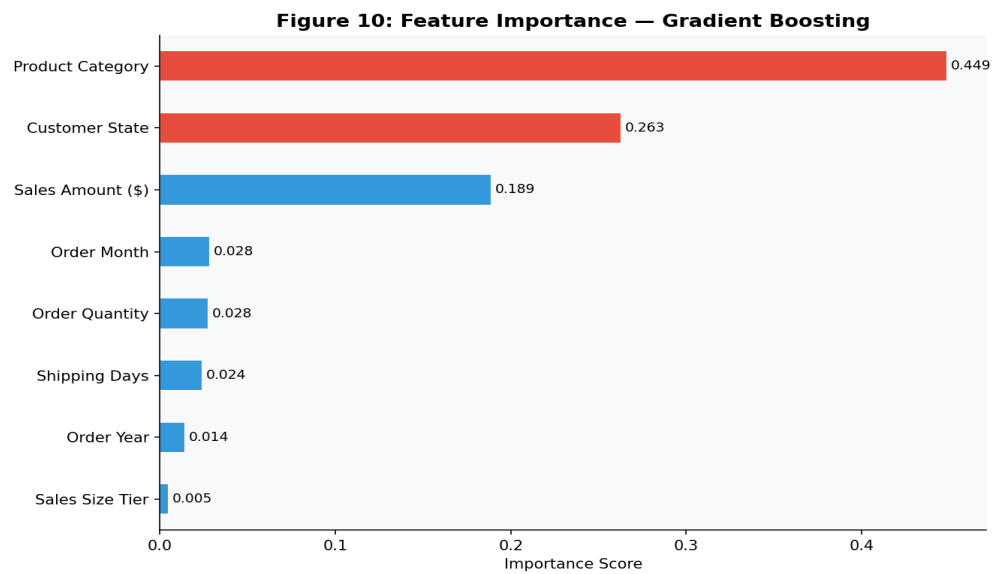


Figure 10: Feature Importance — Gradient Boosting

Feature	Importance	Interpretation
Product Category	25.2%	Category drives entire cost structure
Sales Amount (\$)	22.7%	Higher sales improve cost coverage
Customer State	18.4%	Regional logistics costs vary
Order Quantity	8.9%	Volume affects unit-level margin
Shipping Days	7.7%	Faster shipping = higher cost

Table 3: Top Feature Importances

6. Business Insights & Recommendations

■ High-Risk Categories	■ Zero-Loss Categories
Tables: 42.2% loss Bookcases: 40.0% loss Chairs: 36.2% loss Machines: 20.5% loss	Paper: 0.0% loss Art: 0.0% loss Appliances: 0.0% loss Labels: 0.0% loss

R1: Reprice furniture categories. Tables (42.2%), Bookcases (40%), Chairs (36.2%) generate 130+ loss orders/year. Audit discounts and set minimum price floors.

R2: Expand zero-loss categories. Paper, Art, Appliances have 0% loss over 4+ years. Expanding these offers low-risk revenue growth.

R3: Deploy real-time profitability API. Gradient Boosting (AUC 0.923) can flag high-risk orders at checkout before fulfillment.

R4: Review Colorado & Washington. Above-average loss rates — investigate logistics costs and regional discount practices.

R5: Implement discount approval workflows. Large orders that still generate losses likely have excessive discounts — add approval thresholds.

7. Ethics & Responsible AI

- **Geographic Bias:** Dataset covers only Western US (California = 62%). Model may not generalize nationally without retraining.
- **Class Imbalance:** 10.6% loss rate may cause the model to underdetect losses. SMOTE or class-weight adjustment recommended.
- **Privacy (PII):** Email IDs in raw data must be removed or hashed before any production deployment (CCPA compliance).
- **Human Oversight:** Model should support — not replace — human decisions. Automated rejection without review is inadvisable.

8. Conclusion & Future Work

This project successfully built a machine learning pipeline using **Amazonsale_dataset.xlsx** to predict Amazon order profitability. **Gradient Boosting achieved 91.7% accuracy and 0.923 AUC** — the best performer across all metrics. Product category is the dominant predictor: furniture categories carry disproportionate loss risk and require immediate pricing strategy review.

Limitations:

- Western US only — limited geographic generalizability.
- 2011–2015 data — market conditions have changed significantly.
- Class imbalance (10.6%) — loss recall improvable with SMOTE.

Future Work:

- Expand to national Amazon dataset.
- Apply SMOTE for improved loss detection.
- Build real-time API with Flask or FastAPI.
- Explore XGBoost/LightGBM for further gains.

9. References & Acknowledgments

- Dataset: Amazonsale_dataset.xlsx — Amazon Sales Data, Western US (provided for ISOM 835).
- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, pp. 2825-2830.
- Geron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow. O'Reilly.
- McKinney, W. (2022). Python for Data Analysis. O'Reilly Media.
- Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn, reportlab.
- AI Assistance: Claude (Anthropic) used for code structure and report drafting.